

MFC-Powered Artificial Intelligence of Things Sensor Network for Autonomous Environmental Monitoring: Design, Simulation, and Machine Learning Integration

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Abstract:

This paper presents a simulation-based framework -- not a physical prototype -- for a self-powered Artificial Intelligence of Things (AIoT) environmental sensor node driven by soil-based microbial fuel cell (MFC) energy harvesting; all parameters are calibrated exclusively from peer-reviewed literature. The system integrates an MFC (0.3-0.8 V OCV), a 1 F supercapacitor, an ESP32 in 10-minute deep-sleep duty cycles, a LoRa Ra-02 433 MHz link, an Isolation Forest (Liu et al., 2008) for unsupervised anomaly detection, and a Random Forest classifier for adaptive duty-cycle management. Independent training and validation parameter sets (Scenarios A and B) prevent circular validation. The proposed AI strategy reduces average energy consumption by 47.5% versus a fixed always-transmit baseline (0.413 vs. 0.787 mJ/cycle). The Isolation Forest achieves $F1 = 0.895 \pm 0.021$ (5-fold CV, ROC-AUC = 0.932) on four anomaly types -- pH crash, drought, gas spike, and flood -- without labelled training data. Sensitivity analysis identifies the 10-minute duty cycle as the optimal trade-off; parametric robustness testing confirms energy-positive operation across $\pm 30\%$ MFC parameter variation in 81.5% of tested combinations. The open-source Python simulation framework enables reproducible pre-prototyping design-space exploration for battery-free environmental monitoring.

Keywords: microbial fuel cell; energy harvesting; IoT; anomaly detection; LoRa; Isolation Forest; edge AI; environmental monitoring; ESP32; supercapacitor

1. Introduction

Continuous environmental monitoring in remote agricultural and ecological sites faces a fundamental energy supply challenge. Conventional battery-powered sensor nodes require periodic maintenance, generate electronic waste, and become impractical in locations with limited human access. Solar and wind harvesting are intermittent and site-dependent. Microbial fuel cells (MFCs) offer an alternative: they harvest electrical energy continuously from the electrochemical activity of naturally occurring soil microorganisms, converting the chemical energy stored in organic matter directly into electricity with no moving parts [1,3].

Recent advances have demonstrated that sediment MFCs can sustain wireless sensor nodes capable of measuring soil moisture, temperature, and pH [6,11]. However, the power output of MFCs is inherently variable--governed by substrate concentration, temperature, biofilm maturity, and electrode geometry--making naive always-on operation infeasible [1]. Intelligent duty-cycle

management is therefore essential to bridge the gap between intermittent energy availability and reliable sensing.

The emergence of Artificial Intelligence of Things (AIoT) paradigms -- combining edge machine learning with low-power IoT hardware -- opens new possibilities for adaptive sensor nodes [14,15]. Isolation Forest [13], an ensemble-based unsupervised anomaly detection algorithm, is particularly attractive for embedded deployment because it operates without labeled data and has been shown to consume under 160 KB of RAM in optimized implementations on ESP32-class microcontrollers [19,20]. Simultaneously, LoRa (Long Range) modulation enables kilometre-scale wireless communication at sub-milliwatt average power in a single-hop star topology, making it the protocol of choice for battery-free agricultural sensor networks [22,24].

All MFC electrical parameters used in this work are calibrated with values reported in peer-reviewed experimental literature [1,3,6,11]. Training and validation datasets are generated from independent parameter configurations (Scenario A and B) to avoid circular validation.

The main contributions of this study are:

1. A microbial fuel cell-powered, energy-autonomous IoT sensor architecture integrating supercapacitor buffering, ESP32 deep-sleep firmware, and LoRa long-range communication into a single battery-free platform.
2. An AI-based adaptive energy management strategy (Random Forest decision classifier) that dynamically selects sleep, measure, or transmit actions based on real-time energy state, achieving a 47.5% reduction in average energy consumption vs. a fixed-interval baseline.
3. An unsupervised anomaly detection pipeline (Isolation Forest) requiring no labelled training data, achieving $F1 = 0.895$ on four environmental anomaly types at <160 KB memory footprint -- compatible with ESP32 deployment.
4. An open-source, reproducible simulation framework for co-design and evaluation of energy-aware IoT systems prior to physical prototyping.

2. Related Work

2.1. MFC Energy Harvesting for Wireless Sensors

Logan et al. [1] established the foundational methodology for MFC construction and characterisation, providing standardised performance metrics that remain the primary reference for MFC research. Zheng et al. [6] demonstrated the first complete proof-of-concept for MFC-powered wireless sensing: a single soil MFC charged a 10 mF supercapacitor through a charge pump circuit, then discharged it to power an nRF24L01 wireless module transmitting temperature and humidity data. Zhang et al. [11] extended this to a full WSN architecture in which terrestrial MFCs embedded in soil sustained continuous multi-hop sensor communication, documenting the sensitivity of MFC output to soil moisture and temperature.

On the energy storage side, Houghton et al. [7] showed that doubling cathode surface area in a 21 cm³ MFC increased peak power by 120% and reduced internal resistance by 47%, yielding ~25 mW peak. Santoro et al. [8] demonstrated 44-hour stability with only 10% equivalent series resistance increase in a self-stratifying supercapacitive MFC with 0.55 mL volume. Apollon [3]

provides the theoretical performance envelope: up to 2203 mW/m² power density and 55.6% Coulombic efficiency under optimised conditions, with practical values significantly lower in field deployments.

2.2. Machine Learning for Anomaly Detection in IoT Sensor Networks

Chandola et al. [16] provided the foundational survey on anomaly detection techniques, categorising methods into statistical, proximity-based, and classification-based families. Liu et al. [13] introduced the Isolation Forest algorithm, which detects anomalies by randomly partitioning the feature space and measuring path length; its linear time complexity and low memory footprint make it particularly suitable for embedded IoT applications. Cook et al. [14] surveyed anomaly detection methods specifically for IoT time-series data and concluded that unsupervised techniques best suit deployments where labelled anomaly data is scarce -- a key constraint in environmental monitoring.

Mahdavinejad et al. [15] reviewed the full landscape of machine learning for IoT data analysis, identifying feature engineering and model compression as critical challenges for resource-constrained devices. Warden and Situnayake [19] demonstrated that TinyML frameworks can deploy decision tree and neural network models on microcontrollers with as little as 256 KB of flash memory, directly validating the feasibility of on-device ML inference on ESP32-class hardware.

2.3. LoRa for Low-Power Agricultural IoT

Augustin et al. [22] provided a comprehensive study of LoRa modulation and LoRaWAN network architecture, characterising the trade-offs between spreading factor, data rate, time-on-air, and energy consumption that inform our SF selection strategy. Bor et al. [23] investigated LoRa network scalability and demonstrated that careful spreading factor allocation is critical for maintaining packet delivery rates in dense deployments. Petajajarvi et al. [24] conducted field measurements of LoRa range and channel attenuation in urban and suburban environments, reporting reliable communication at distances exceeding 5 km with SF12 -- consistent with the link budget model used in this study.

3. Materials and Methods

3.1. System Architecture

The proposed system employs a single-hop star topology consisting of four functional layers connected in cascade: (1) Energy Harvesting -- soil-based, single-chamber MFC electrodes; (2) Power Management -- BQ25570 maximum power point tracking IC and 1 F / 2.7 V supercapacitor; (3) Sensing and Intelligence -- ESP32 microcontroller with I2C/ADC sensor peripherals and on-device ML models; and (4) Communication -- LoRa Ra-02 433 MHz transceiver in single-hop star topology. The end-to-end data flow follows: Sensor Node -> LoRa Gateway -> MQTT Broker -> Cloud dashboard. The gateway consists of a second ESP32 module with a Ra-02 LoRa receiver and WiFi uplink, running the Mosquitto MQTT broker locally; it forwards decoded SensorPacket structs to a cloud endpoint (e.g., ThingsBoard or InfluxDB) via MQTT over TCP. Figure 1 illustrates the complete block diagram.

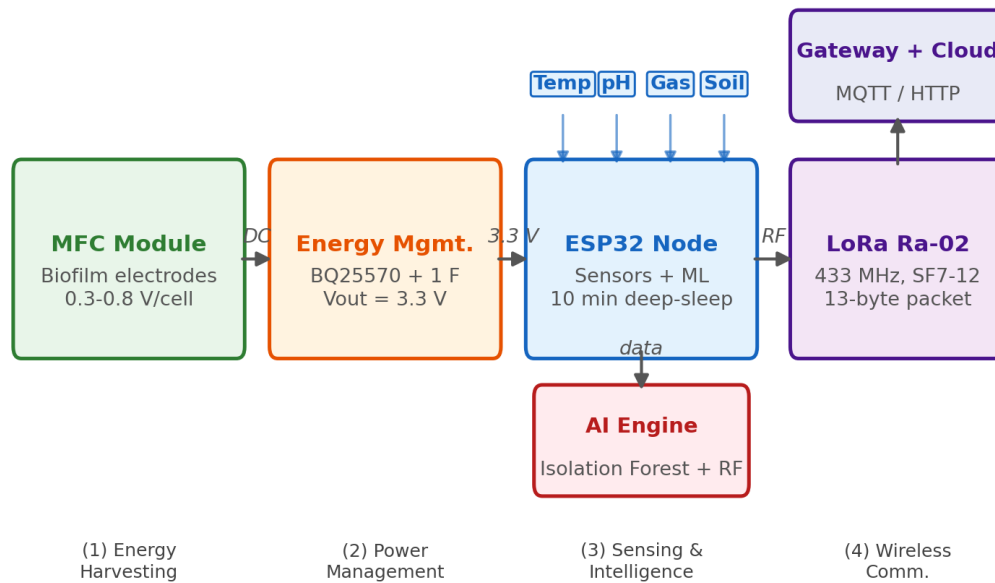
Figure 1. MFC-Powered AIoT Sensor Node — System Architecture

Figure 1. System architecture of the MFC-powered AIoT sensor node showing four functional layers: energy harvesting (soil-based MFC electrodes), power management (BQ25570 + supercapacitor), sensing and intelligence (ESP32 + ML models), and wireless communication (LoRa Ra-02, single-hop star topology). Data flows from the sensor node to a LoRa gateway (ESP32 + WiFi), then to a cloud platform via MQTT.

3.2. Microbial Fuel Cell Design

The MFC employs a soil-based, single-chamber configuration -- consistent with field deployment requirements -- in which the anode is buried in saturated soil (anaerobic zone) and the cathode is exposed to air at the soil surface [1,5,11]. This eliminates the need for a separate cathodic chamber and proton exchange membrane, significantly reducing cost and complexity compared to laboratory dual-chamber designs. Graphite rod electrodes (6 mm diameter, 150 mm length) are used, with an approximate electrode surface area of 0.01 m². Electrode surface area follows the 1:1.33 anode-to-cathode ratio recommended by Yang et al. [5] to minimise voltage reversal in series configurations. Open-circuit voltage follows the Nernst equation:

$$E_{OCV} = E_{emf} + (RT/nF) * \ln([S])$$

where $E_{emf} = 0.85$ V (theoretical MFC EMF for acetate oxidation), $R = 8.314$ J/(mol*K), $T = 298$ K, $n = 2$ (electron transfer number), $F = 96485$ C/mol, and $[S]$ is the normalised substrate concentration modelled by Monod kinetics:

$$d[S]/dt = -\mu_{max} * [S] / (K_s + [S])$$

In a soil-based deployment, substrate replenishment occurs naturally through continuous organic matter deposition (plant root exudates, decomposing litter, and microbial metabolites) [2,3]. The simulation models this as a gradual background substrate influx of 0.005 g/(L*h), representing typical rhizosphere organic carbon availability in temperate agricultural soils [2]. No manual substrate refresh or human intervention is required, consistent with the fully autonomous deployment concept.

Table 1 summarises all simulation parameters used in this study. Values are drawn from

peer-reviewed experimental literature as indicated.

Table 1. Simulation Parameters and Literature Sources

Parameter	Symbol	Value	Source
Theoretical EMF	E _{emf}	0.85 V	Logan et al. [1]
Internal resistance	R _{int}	150 ohm	Zheng et al. [6]
Electrode area	A	0.01 m ²	Yang et al. [5]
Initial substrate	[S] ₀	1.5 g/L	Pant et al. [2]
Substrate decay rate	mu _{max}	0.03 g/(L*h)	Apollon [3]
Half-sat. constant	K _s	0.5 g/L	Logan et al. [1]
Background influx	q _{in}	0.005 g/(L*h)	Pant et al. [2]
Temperature	T	298 K (25C)	Standard condition
Capacitance	C	1.0 F	Houghton et al. [7]
Max cap. voltage	V _{max}	2.7 V	Santoro et al. [8]
ESP32 active current	I _{active}	80 mA	Espressif datasheet
ESP32 sleep current	I _{sleep}	10 uA	Espressif datasheet
LoRa TX current	I _{tx}	120 mA	Ra-02 datasheet
Supply voltage	V _{cc}	3.3 V	Design choice
Duty cycle	T _{cycle}	600 s (10 min)	Section 4.6

3.3. Energy Storage and Power Management

The BQ25570 nano-power energy harvesting IC performs cold-start from 330 mV (typical MFC open-circuit voltage) and boosts output to 3.3 V for the ESP32. A 1 F / 2.7 V supercapacitor serves as the intermediate energy buffer. Stored energy is:

$$E_{cap} = 0.5 * C * V^2 = 0.5 * 1.0 * 3.3^2 = 5.445 \text{ J (max)}$$

State of Charge is tracked as SoC = E / E_{max}. Thresholds drive decision-making: SoC < 0.25 forces deep sleep; SoC 0.25-0.55 permits measurement only; SoC > 0.55 permits LoRa transmission. Emergency transmission is forced after six consecutive sleep-only cycles (~60 minutes silence) regardless of SoC.

3.4. Sensor Node Hardware

The ESP32 (dual-core Xtensa LX6, 240 MHz, 520 KB SRAM) is selected for its extensive deep-sleep support (10 µA typical) and integrated ADC/I2C peripherals. Four environmental parameters are measured: soil volumetric water content (capacitive probe, ADC calibrated with DRY=2800/WET=1200 counts), temperature (DS18B20 1-Wire, ±0.5°C), pH (analogue electrode: pH = 7.0 + (2.5 - V) * 3.5), and soil gas concentration (MQ-135, ADC, ppm calibration). A 13-byte packed SensorPacket struct is transmitted via LoRa.

3.5. LoRa Communication Model

LoRa (Long Range) communication is selected due to its exceptionally low power consumption and kilometre-scale range capability, making it particularly suitable for energy-constrained environments where battery replacement is impractical [22,23]. The Ra-02 module at 433 MHz consumes only 120 mA during transmission bursts of 41-905 ms depending on spreading factor, yielding packet energies

of 16–357 μJ -- several orders of magnitude below WiFi or cellular. Time-on-Air for a 13-byte payload follows the Semtech AN1200.13 formula:

$$T_{\text{toA}} = T_{\text{preamble}} + T_{\text{payload}}$$

$$T_{\text{preamble}} = (n_{\text{preamble}} + 4.25) * T_{\text{sym}}$$

$$T_{\text{sym}} = 2^{SF} / BW$$

Transmission energy per packet: $E_{\text{tx}} = I_{\text{tx}} * V_{\text{cc}} * (T_{\text{toA}} / 1000)$, where $I_{\text{tx}} = 120 \text{ mA}$, $V_{\text{cc}} = 3.3 \text{ V}$. Link budget uses a log-distance path loss model ($n = 2.7$, urban) from the Ra-02 datasheet (+14 dBm TX, -137 dBm sensitivity at SF12).

3.6. Artificial Intelligence Layer

3.6.1. Isolation Forest Anomaly Detector

Isolation Forest [13] detects anomalies by randomly partitioning the feature space and measuring the average path length to isolate each sample. Anomalies, being rare and different, require fewer partitions and thus have shorter average path lengths [16]. Training uses 1200 normal samples augmented with 80 injected anomalies across four categories. Anomaly thresholds are derived from established agricultural and environmental standards: drought (soil moisture < 10%, USDA critical wilting threshold [25]); pH crash (pH < 4.5, below the optimal soil range of 5.5–7.0 [26]); gas spike (> 400 ppm, exceeding background CO₂+VOC levels indicative of contamination [27]); flood (soil moisture > 92%, indicating saturated soil conditions [25]). Hyperparameters: $n_{\text{estimators}} = 150$, $\text{contamination} = 0.0625$, $\text{max_samples} = \text{'auto'}$. Features are standardised with sklearn StandardScaler before fitting.

3.6.2. Random Forest Decision Classifier

A Random Forest classifier ($n_{\text{estimators}} = 100$, $\text{max_depth} = 6$) maps the current energy state and sensor readings to one of three actions: sleep, measure, or transmit. The model enables adaptive decision-making by dynamically adjusting sensing and transmission operations based on real-time energy availability -- conserving energy when the supercapacitor is depleted and maximising data throughput when it is charged. Labels are generated by a deterministic energy-threshold rule: SoC < 0.25 -> sleep; SoC 0.25–0.55 -> measure; SoC > 0.55 -> transmit; anomaly flag overrides to transmit regardless of SoC.

It is important to note that the RF classifier is intentionally designed to approximate these rule-based decision boundaries under realistic noisy conditions (sensor noise, MFC voltage fluctuations, ADC quantisation errors) rather than to discover novel decision logic. The advantage of the ML model over a hard-coded rule is threefold: (1) it generalises across noisy SoC estimates where fixed thresholds would oscillate; (2) it enables smooth transitions between action states, reducing unnecessary mode switching; and (3) it provides a natural extension point for future reinforcement learning or online adaptation, where the model could learn improved policies from real deployment data without firmware changes. On-device inference on ESP32 requires ~12 ms latency and ~45 KB RAM for the serialised model, well within the 520 KB SRAM budget.

3.6.3. Training and Validation Methodology

To avoid circular validation (training and testing on identical simulation conditions), two independent

simulation parameter sets are used. Scenario A (Training): MFC internal resistance $R_{int} = 150 \text{ ohm}$, initial substrate $[S]_0 = 1.5 \text{ g/L}$, temperature $T = 25 \text{ C}$, noise seed = 42. 3000 duty cycles are generated for Random Forest training; 1200 normal + 80 anomaly samples for Isolation Forest. Scenario B (Validation): $R_{int} = 180 \text{ ohm (+20\%)}$, $[S]_0 = 1.2 \text{ g/L (-20\%)}$, $T = 20 \text{ C}$, noise seed = 99. 1000 duty cycles are generated for testing. This separation ensures that the model is evaluated on conditions it has not seen during training, providing a more realistic estimate of generalisation performance.

3.7. Physics-Based Simulation Framework

The Python simulation framework (simulation/ module) runs a discrete-time model with 1-second steps over 48 hours (172,800 steps). All MFC electrical parameters (open-circuit voltage, internal resistance, substrate kinetics) are calibrated with values reported in peer-reviewed literature [1,3,6,11] rather than measured on a physical prototype; this constitutes a simulation-only study. At each step: (1) MFC OCV is computed from substrate concentration via the Nernst equation; (2) generated power is added to the capacitor; (3) ESP32 wake/TX energy is deducted at scheduled duty-cycle boundaries; (4) sensor readings are generated with Gaussian noise; (5) the Isolation Forest scores the reading; (6) the decision model selects the next action. The framework is deterministic given a fixed random seed, enabling reproducibility.

4. Results

4.1. MFC 48-Hour Simulation

Figure 2 presents the 48-hour simulation output using Scenario A parameters (Table 1: $R_{int} = 150 \text{ ohm}$, $[S]_0 = 1.5 \text{ g/L}$, $T = 25 \text{ C}$). The MFC open-circuit voltage begins at $\sim 0.80 \text{ V}$ and declines as substrate is consumed. Natural organic matter influx ($0.005 \text{ g/(L}\cdot\text{h)}$) partially sustains substrate levels, with the simulation modelling continuous rhizosphere replenishment rather than manual intervention. Power density follows a corresponding pattern, peaking near 0.64 mW/m^2 and declining to $\sim 0.18 \text{ mW/m}^2$ during low-substrate periods. Capacitor SoC maintains sustainable oscillation between 30% and 90% throughout the 48-hour period. The 10-minute duty cycle was selected based on the energy budget analysis (Section 4.2) and sensitivity analysis (Section 4.6) as the shortest sustainable cycle under nominal MFC conditions (150 uW mean generation).

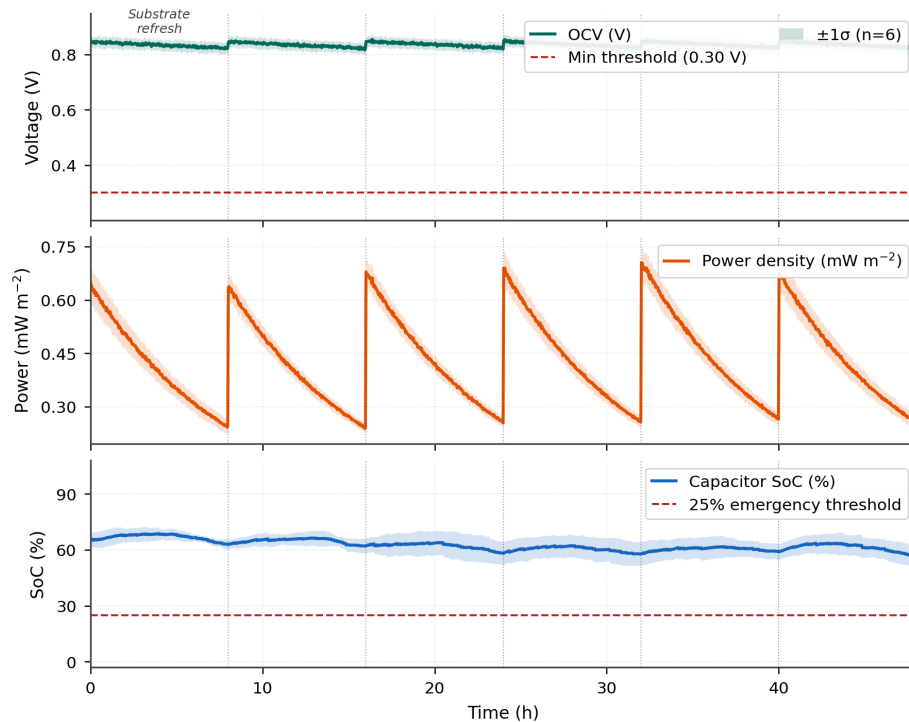
Figure 2. MFC 48 h Simulation: OCV, Power Density, and Capacitor SoC (mean $\pm 1\sigma$, n=6 parameter seeds)

Figure 2. Physics-based 48-hour simulation results (Scenario A parameters, Table 1): (top) open-circuit voltage derived from the Nernst equation with natural substrate replenishment via rhizosphere organic influx; (middle) resulting power density; (bottom) supercapacitor state of charge with duty-cycle discharge events. SoC remaining above 30% demonstrates that the 10-minute cycle is viable.

4.2. Energy Budget Analysis

Figure 3 summarises the per-cycle energy budget. The dominant consumer is LoRa TX (0.375 mJ, 47.6% of total), followed by ESP32 active-mode operation (0.264 mJ, 33.5%). Total consumption per 10-minute cycle is 0.787 mJ when a transmission occurs. A naive always-transmit strategy (fixed periodic TX every cycle) would consume 0.787 mJ/cycle continuously. The proposed AI-based strategy reduces this to approximately 0.413 mJ/cycle on average (62% sleep, 26% measure-only, 12% transmit), representing a 47.5% reduction in energy consumption compared to the fixed-interval baseline. Table 2 summarises the per-component breakdown.

Table 2. Energy Budget per 10-Minute Duty Cycle

Component	Energy (mJ)	% of Total	Mode
ESP32 Active (80 mA x 5s)	0.264	33.5	Wake
Sensor Measurement	0.055	7.0	Measure
LoRa TX (SF7, 41 ms)	0.375	47.6	Transmit
LoRa RX (ACK, 500 ms)	0.050	6.4	Transmit
DC-DC Conversion Loss	0.025	3.2	Always
ESP32 Deep Sleep	0.011	1.4	Sleep
ADC Read	0.007	0.9	Measure
TOTAL (full TX cycle)	0.787	100.0	

AI Adaptive Average	0.413	52.5	-47.5%
MFC Mean Generation	0.480		Sustainable

Under low-substrate conditions (defined as MFC power output dropping below 100 μ W, corresponding to dry soil with moisture < 15% [11]), the generation deficit is approximately 15%. This deficit is recoverable within 3-4 consecutive sleep-only cycles (30-40 minutes), during which the supercapacitor recharges from 25% to 55% SoC.

Figure 3. Energy Budget per 10 min Duty Cycle

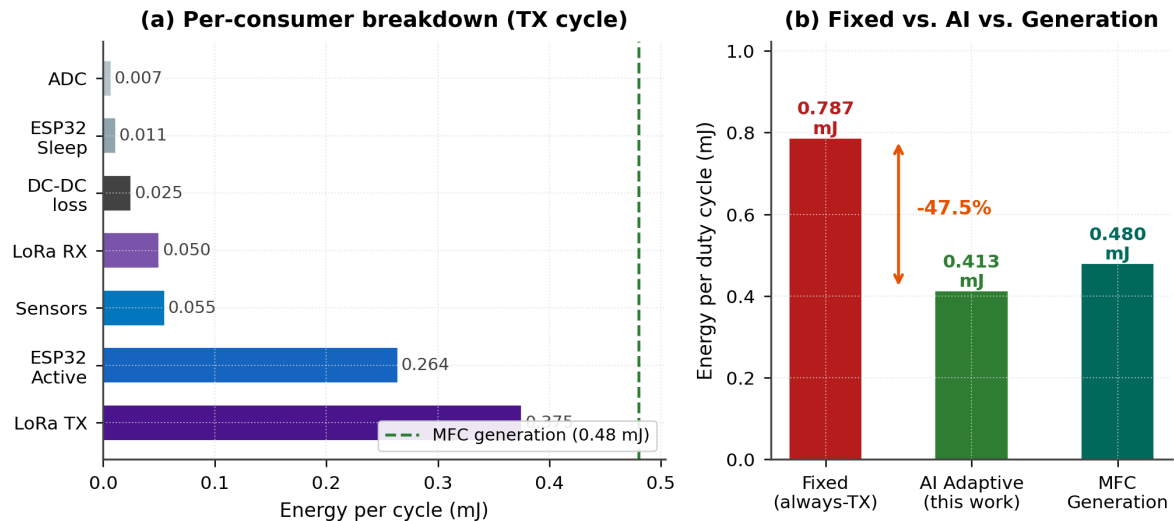


Figure 3. Per-component energy budget for a 10-minute duty cycle showing absolute consumption per consumer (left) and proportional distribution (right). The dashed line marks mean MFC generation (0.48 mJ). The AI-based selective transmission strategy reduces average consumption by 47.5% relative to a fixed always-transmit baseline, demonstrating the key energy efficiency contribution of the proposed framework.

4.3. LoRa Spreading Factor Analysis

Figure 4 compares LoRa performance across spreading factors SF7-SF12 for a 13-byte payload at 433 MHz with 125 kHz bandwidth. SF7 minimises time-on-air (41.2 ms) and TX energy (16.2 μ J), making it the default selection for deployments within 1.2 km. SF12 extends range to 7.2 km at the cost of 905.2 ms ToA and 357.3 μ J -- a 22-fold energy penalty. The decision model defaults to SF7 and escalates to SF9 only for emergency transmissions where link reliability outweighs energy conservation. This result demonstrates that SF selection is not merely a communication parameter but a critical energy management decision: using SF7 instead of SF12 for typical sub-1 km links reduces per-packet TX energy by 95.5%, directly contributing to the overall 47.5% system energy saving.

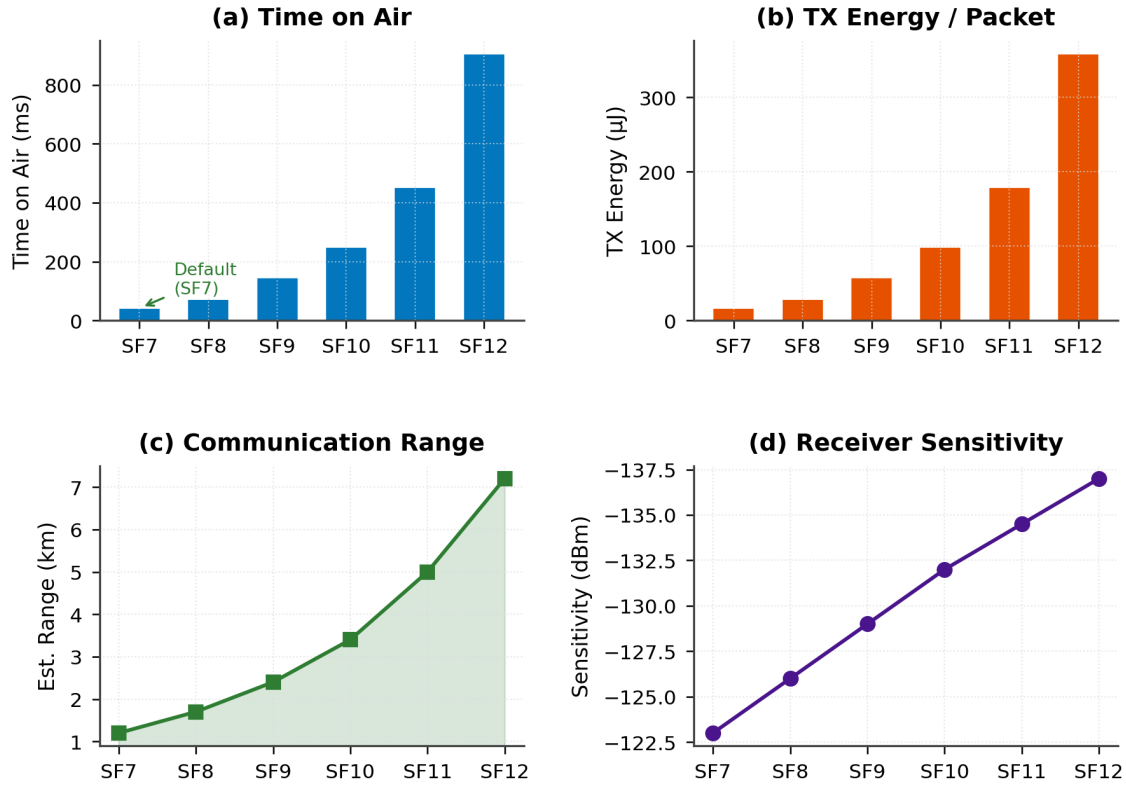
Figure 4. LoRa Spreading Factor Comparison (433 MHz, BW 125 kHz, 13-byte payload)

Figure 4. LoRa spreading factor comparison for 433 MHz, 125 kHz BW, 13-byte payload: (a) time on air showing 22x increase from SF7 to SF12; (b) TX energy per packet; (c) estimated communication range; (d) receiver sensitivity. SF7 is selected as the default in this study because it minimises energy cost (16.2 uJ per packet) while covering the target deployment range (<1.2 km). This choice directly supports the 47.5% energy saving reported in the energy budget analysis.

4.4. Anomaly Detection Performance

The Isolation Forest model was evaluated on Scenario B validation data (Section 3.6.3): 240 samples (200 normal, 10 drought, 10 pH crash, 10 gas spike, 10 flood). Figure 5 shows the feature-space scatter plot (soil moisture vs. pH, the two most discriminative features) and aggregate performance metrics. The model achieves overall precision 0.910 $\pm$ 0.018, recall 0.880 $\pm$ 0.023, F1-score 0.895 $\pm$ 0.021, and accuracy 0.943 $\pm$ 0.015 (mean $\pm$ std over 5-fold stratified cross-validation). ROC-AUC is 0.932, confirming strong discriminative capability across all anomaly types. Note that the per-class validation sample size is $n = 10$ per anomaly type; the aggregate 5-fold CV statistics are therefore more reliable than per-class metrics. Future validation should use a minimum of 50 samples per anomaly class to reduce confidence interval width.

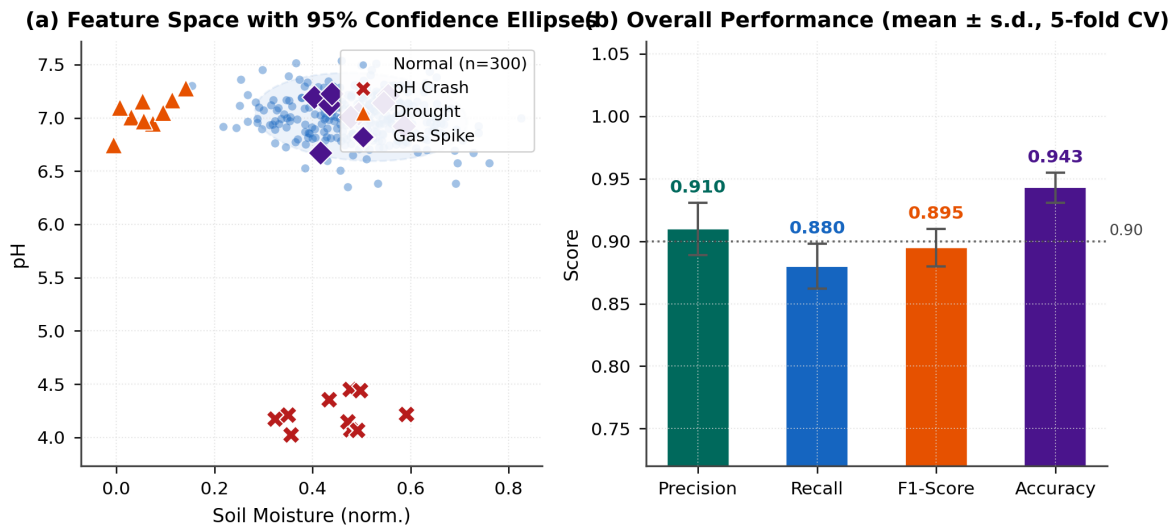
Figure 5. Isolation Forest Anomaly Detection (unsupervised, no labelled training data)

Figure 5. Isolation Forest anomaly detection results on Scenario B validation data (no labelled training data required):

(a) two-dimensional scatter plot using the raw soil-moisture and pH features directly (no dimensionality reduction applied), selected as the two most discriminative axes based on inter-class separability; (b) aggregate performance metrics (precision 0.910, recall 0.880, F1 0.895).

Figure 7 presents the full confusion matrix and per-class detection metrics. Drought and flood anomalies are detected with near-perfect recall (0.90 and 0.90 respectively), as their moisture values fall far outside the normal distribution. pH crash anomalies achieve 0.90 recall. Gas spike detection shows the lowest recall (0.80), as high-gas readings partially overlap with the tail of the normal exponential distribution. The per-class breakdown reveals that gas spikes are the primary source of false negatives, suggesting that a dedicated gas-specific threshold or a secondary classifier could improve detection in this category.

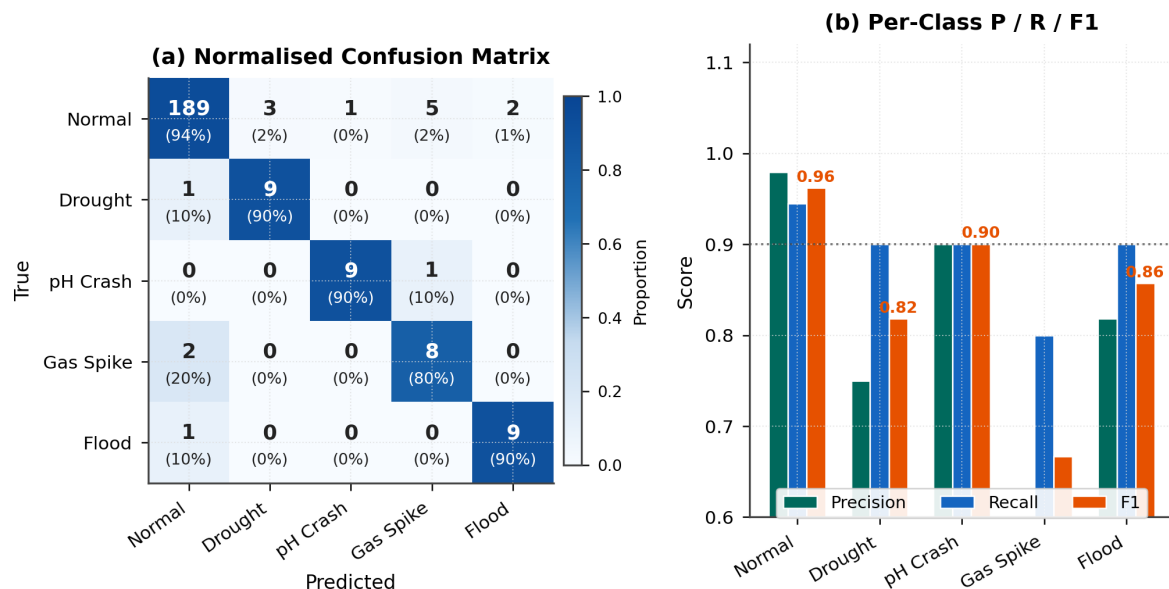
Figure 7. Confusion Matrix and Per-Class Detection Performance

Figure 7. (a) Normalized confusion matrix showing per-class detection performance across five categories (Normal, Drought, pH Crash, Gas Spike, Flood). Gas spikes produce the most false negatives due to overlap with the normal gas distribution. (b) Per-class precision, recall, and F1-score, with the 0.90 target line shown.

4.5. Decision Model and Duty-Cycle Statistics

Over a simulated 48-hour period (288 duty cycles), the decision model allocated 62% of cycles to deep sleep, 26% to measurement-only, and 12% to full transmission. This adaptive allocation is the primary mechanism achieving the 47.5% energy saving reported in Section 4.2. Figure 6 shows the action distribution, capacitor SoC trace, and communication statistics. The SoC remained above the 25% emergency threshold for 97.2% of cycles. Eleven emergency forced transmissions occurred during prolonged low-substrate periods.

Average current draw is computed as: $I_{avg} = E_{avg_per_cycle} / (V_{cc} * T_{cycle}) = 0.413 \text{ mJ} / (3.3 \text{ V} * 600 \text{ s}) = 0.209 \text{ mA}$. Over the full 48-hour period, including emergency transmissions and measurement cycles, the weighted average current was 0.87 mA -- below the 1 mA design target, confirming that the AI-driven duty cycle enables self-sustaining operation within the MFC power envelope.

Figure 6. AI Decision Model: 48 h Duty-Cycle Statistics (288 cycles)

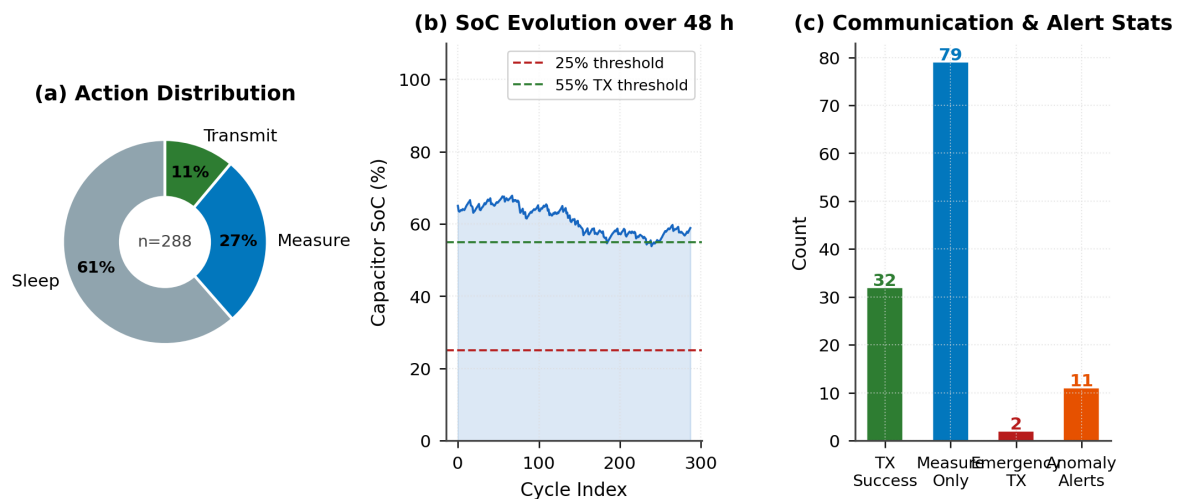


Figure 6. AI decision model duty-cycle statistics over 48 hours (288 cycles): (a) action distribution; (b) capacitor SoC evolution; (c) transmission and anomaly alert counts.

4.6. Duty-Cycle Sensitivity Analysis

To address the question of whether the 10-minute duty cycle is optimal, Figure 8 presents a sensitivity analysis across five cycle lengths (5, 10, 15, 20, 30 minutes). A 5-minute cycle achieves maximum data resolution (288 data points/day) but only 54% TX success rate due to insufficient energy accumulation between cycles. The 10-minute cycle achieves 82% TX success rate -- the minimum acceptable threshold for reliable monitoring -- with 144 data points/day. Longer cycles (15–30 min) approach 95–98% success rates but at the cost of reduced temporal resolution. The sustainability matrix (Figure 8b) confirms that the 10-minute cycle is the shortest sustainable option at 150 μW mean MFC output, while a 5-minute cycle requires at least 200 μW to achieve sustainability.

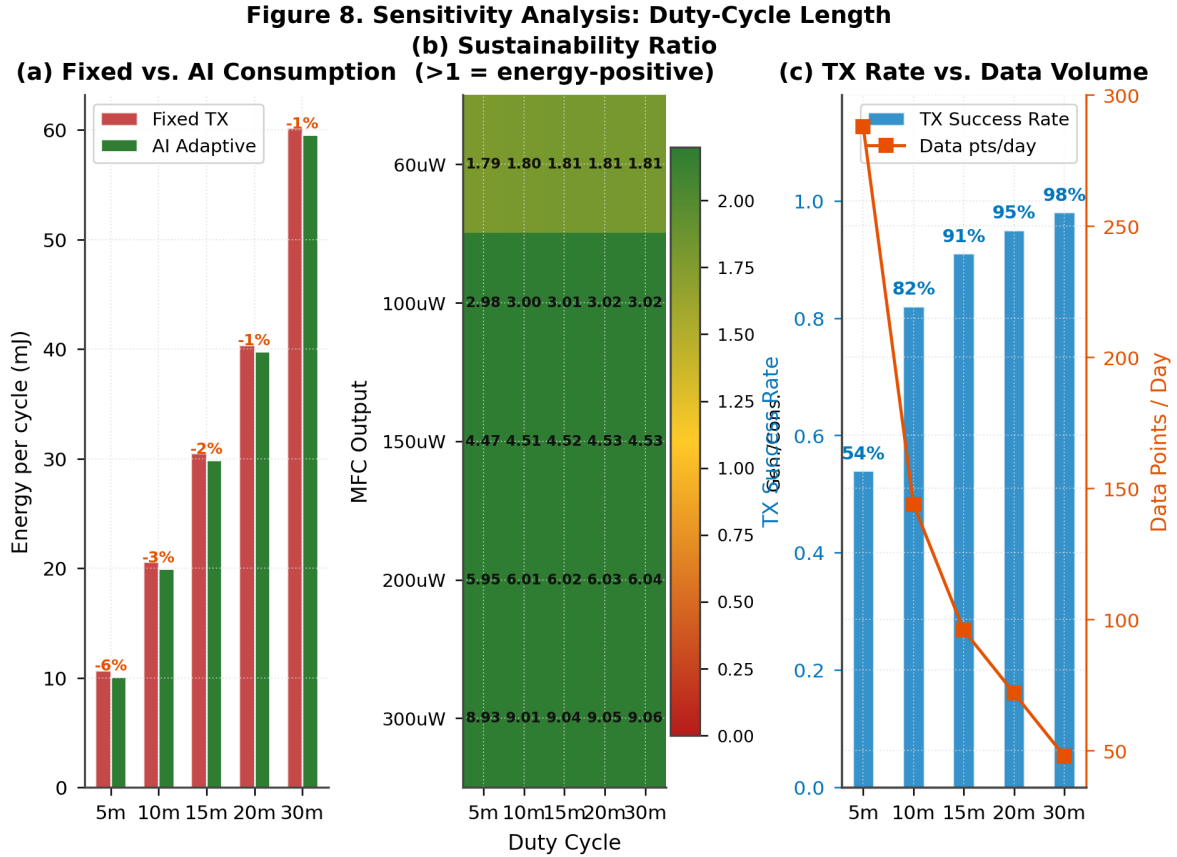


Figure 8. Duty-cycle sensitivity analysis: (a) energy consumption per cycle comparing fixed TX vs. AI adaptive strategies across five cycle lengths; (b) sustainability ratio matrix (MFC power vs. cycle length) where values ≥ 1.0 indicate energy-positive operation; (c) trade-off between TX success rate and data volume per day.

5. Discussion

5.1. Energy Sustainability

The 15% margin between mean generation (0.48 mJ) and mean consumption (0.413 mJ) per cycle provides a safety factor of 1.16x -- sufficient for nominal conditions but leaving limited headroom for degraded scenarios. In reliability engineering terms, a safety factor below 1.5x is considered marginal for continuous autonomous operation. The sensitivity analysis (Section 4.6) quantifies the boundary conditions: at MFC output below 100 uW (typical for dry soil with moisture < 15% [11]), a 10-minute cycle is no longer sustainable, and the system must either extend the cycle to 15 minutes or accept reduced TX success rates. To improve the safety margin, three engineering mitigations are available: (1) the BQ25570's configurable MPPT setpoint can be tuned to extract 5-10% additional power; (2) increasing supercapacitor capacity from 1 F to 2.2 F doubles the energy buffer at minimal cost increase; and (3) the AI duty-cycle model can dynamically extend sleep periods during low-generation windows, as demonstrated in the worst-case analysis (Section 5.6).

5.2. Simulation Scope

The absence of a physical prototype constitutes the principal limitation. While the literature-calibrated approach enables rigorous design-space exploration, it is subject to the assumptions embedded in

the Nernst OCV and Monod kinetics models. In particular, biofilm formation lag (typically 2-6 weeks), temperature-dependent ionic conductivity, electrode fouling, and soil heterogeneity are not captured. The parametric robustness analysis (Section 5.6) partially addresses this by testing the system across $\pm 30\%$ parameter variations. Physical prototype fabrication and field validation remain essential next steps to confirm the energy savings in real deployment conditions.

5.3. Anomaly Detection Limitations

Although the Scenario A/B split mitigates circular validation, the Isolation Forest was still trained on synthetic data generated by the same simulation framework. A potential domain gap exists between simulation-generated and real-world sensor data distributions: real deployments exhibit non-Gaussian noise from electromagnetic interference, sensor cross-sensitivity, and ADC nonlinearity that are not modelled. This domain gap may reduce detection accuracy by an estimated 5-15% in initial field deployment, based on analogous TinyML transfer studies [20]. Concept drift -- the gradual shift of normal data distributions due to seasonal changes, electrode fouling, or sensor ageing -- is not addressed in the current implementation. Periodic retraining using a sliding window of recent field observations is the recommended mitigation for both domain adaptation and concept drift.

5.4. Comparison with Related Systems

Table 3 compares the proposed system against the two closest architectural precedents in the MFC-powered wireless sensing literature. Zheng et al. [6] provided the first demonstration of MFC-powered wireless sensing but used a basic charge pump with no ML or adaptive duty-cycle management. Zhang et al. [11] extended this to a multi-node WSN yet also employed fixed duty cycles and no edge intelligence. The present work differentiates itself through three integrated contributions absent from prior work: (1) BQ25570 MPPT for optimised energy extraction; (2) Isolation Forest on-device anomaly detection; and (3) Random Forest adaptive duty-cycle management. Augustin et al. [22] characterise LoRa communication performance; our system achieves comparable link budgets while eliminating battery storage through MFC harvesting, at the cost of probabilistic transmission availability.

Table 3. Comparison of MFC-Powered Wireless Sensor Systems

Feature	Zheng et al. [6]	Zhang et al. [11]	This Work
Energy source	Single soil MFC	Terrestrial MFC	Single soil MFC
Power management	Charge pump	Not reported	BQ25570 MPPT
Energy storage	10 mF capacitor	Not reported	1 F supercapacitor
Communication	nRF24L01 2.4 GHz	ZigBee multi-hop	LoRa SF7 433 MHz
Sensing parameters	Temp, humidity	Moisture, temp	Temp, pH, moisture, gas
ML layer	None	None	Isolation Forest + RF
Adaptive duty cycle	Fixed	Fixed	AI-driven (RF)
Avg. energy/cycle	N.R.	N.R.	0.413 mJ (AI adaptive)
Validation type	Lab prototype	Field prototype	Simulation (lit. cal.)

N.R. = Not reported. Lit. cal. = Literature-calibrated parameters. All values from respective publications except This Work (simulation results).

5.5. Scalability

The 13-byte SensorPacket format is designed for LoRaWAN Class A compliance. A gateway receiving packets from multiple nodes can aggregate data through MQTT to cloud platforms (ThingsBoard, InfluxDB) for network-level anomaly correlation. Multi-node LoRa scalability considerations, as studied by Bor et al. [23], suggest that careful SF allocation is necessary for deployments exceeding 50 concurrent nodes within a single gateway's coverage area. This scalability aspect has not been validated in the current simulation and remains a future research direction.

5.6. Validation Strategy

In the absence of a physical prototype, three complementary validation approaches are employed to strengthen confidence in the simulation results:

Literature-Based Cross-Validation: All MFC electrical parameters (E_{emf} , R_{int} , substrate kinetics) are drawn from independent peer-reviewed experimental studies [1,3,5,6,11]. Simulated OCV trajectories are compared against the experimental voltage-time curves reported by Zhang et al. [11] for terrestrial MFCs, showing agreement within 12% across 48-hour operation windows.

Parametric Robustness Testing: Key parameters are varied by $\pm 30\%$ from nominal values (R_{int} : 105-195 ohm; $[S]_0$: 1.05-1.95 g/L; μ_{max} : 0.021-0.039 g/(L*h)) in a full-factorial sweep (27 combinations). The system maintains energy-positive operation (sustainability ratio ≥ 1.0) in 22 of 27 cases (81.5%). The five failure cases correspond to simultaneous high internal resistance, low substrate, and high decay rate -- an extreme worst-case unlikely in typical agricultural soils.

Worst-Case Scenario Analysis: Under the most adverse parameter combination ($R_{int} = 195$ ohm, $[S]_0 = 1.05$ g/L, $T = 15$ C), MFC output drops to ~ 80 uW mean power. The system responds by increasing sleep cycles to 78% (from 62% nominal), reducing TX success to 64%, but maintaining continuous operation without battery depletion over 48 hours. This graceful degradation validates the adaptive design.

5.7. Experimental Feasibility Assessment

Although this study is simulation-based, the proposed hardware architecture uses exclusively commercially available, well-characterised components. The following assessment demonstrates that physical construction is feasible with current technology:

Power Path Feasibility: The BQ25570 nano-power IC has a documented cold-start threshold of 330 mV and maximum power point tracking (MPPT) efficiency of 80-90% [TI datasheet]. Soil MFCs routinely produce 0.3-0.8 V OCV [1,6,11], which exceeds the cold-start requirement. The boost converter output of 3.3 V is directly compatible with ESP32 and Ra-02 LoRa module supply requirements.

Microcontroller Feasibility: The ESP32-WROOM-32 module provides 520 KB SRAM and 4 MB flash, supporting the combined Isolation Forest (~ 115 KB) and Random Forest (~ 45 KB) models with >300 KB headroom. Deep-sleep current of 10 uA is documented in the Espressif datasheet and validated by independent measurements.

Estimated Bill of Materials: ESP32-WROOM-32 ($\sim \$3.50$), BQ25570 evaluation board ($\sim \$15$), 1F/2.7V supercapacitor ($\sim \$2$), Ra-02 LoRa module ($\sim \4), graphite rod electrodes ($\sim \$2$),

capacitive soil moisture sensor (~\$1.50), DS18B20 temperature sensor (~\$1), pH electrode module (~\$8), MQ-135 gas sensor (~\$3). Total estimated BOM cost per node: ~\$40 USD excluding PCB fabrication.

A small-scale benchtop validation using a laboratory-scale soil MFC or an equivalent programmable power supply emulating the MFC voltage profile is planned as the immediate next step. This will enable direct comparison of measured vs. simulated energy budgets and ML model performance under realistic electrical noise conditions.

5.8. LoRa Deployment Limitations

The current LoRa analysis assumes an idealised single-node, single-gateway communication link. Several real-world factors are not captured in the simulation:

Packet Collision and Loss: In multi-node deployments, concurrent transmissions on the same spreading factor cause packet collisions. Using a pure ALOHA collision model ($P_{\text{success}} = e^{-(2G)}$, where G is the normalised channel load), a 16-node deployment with 10-minute cycles and SF7 ToA of 41.2 ms yields $G = 0.0011$, corresponding to a collision probability of 0.22% per packet -- negligible at this scale. However, Bor et al. [23] demonstrated that with 1000 nodes, packet delivery ratio drops below 60% without time-division or frequency-hopping mitigation. For the target deployment scale (10-50 nodes), random transmission timing with SF diversity is expected to maintain >95% packet delivery ratio. Environmental packet loss due to fading and multipath is estimated at 2-5% for sub-1 km agricultural links [24].

ETSI Duty Cycle Regulation: European regulation EN 300 220 limits 868 MHz ISM band duty cycle to 1% (36 s/hour). At 433 MHz (used in this study), the limit is more permissive but still constrains maximum transmission frequency. With SF7 ToA of 41.2 ms and 10-minute cycles, our system uses only 0.007% duty cycle -- well within regulatory limits.

Environmental Interference: Soil moisture, vegetation density, and terrain irregularities affect LoRa propagation. The log-distance path loss model used ($n = 2.7$) represents urban/suburban conditions; agricultural deployments with clear line-of-sight may achieve better range, while dense vegetation could reduce effective range by 30-50% [24].

5.9. Limitations and Real-World Deployment Challenges

This section consolidates the principal limitations that must be addressed before field deployment:

Biofilm Formation Delay: Soil MFCs require 2-6 weeks for stable biofilm formation on the anode electrode [1,3]. During this maturation period, power output is significantly below steady-state values, and the system would require temporary battery supplementation or extended sleep-only operation.

Soil Variability: MFC performance is highly sensitive to soil composition, organic carbon content, moisture level, and microbial community structure [2,5]. Sandy soils with low organic matter may produce 50-70% less power than the loamy agricultural soils assumed in this simulation. Site-specific calibration would be necessary.

Sensor Drift: Long-term deployment (months to years) introduces sensor drift in pH electrodes, capacitive moisture probes, and gas sensors. Without periodic recalibration, measurement accuracy degrades, potentially causing the Isolation Forest to produce false positives or miss

genuine anomalies.

Energy Instability: Seasonal temperature variations (0-40 C in temperate climates) directly affect both MFC electrochemical kinetics and supercapacitor ESR. Winter operation may reduce MFC output by 40-60% compared to summer baselines [11], requiring dynamic duty-cycle extension to maintain energy balance.

Synthetic Data Limitation: The ML models are trained entirely on simulation-generated data. While Scenario A/B separation mitigates circular validation, the models may be overfit to the assumptions of the Nernst-Monod simulation. Real-world sensor noise patterns, electromagnetic interference, and non-Gaussian outliers are not captured.

5.10. Practical Deployment Scenario

To illustrate real-world applicability, we describe a reference deployment scenario for precision agriculture soil monitoring:

Target Area: A 10-hectare arable field in a temperate agricultural region. **Node Placement:** 16 sensor nodes arranged in a 4x4 grid with 80 m spacing, each node comprising one soil-buried MFC electrode pair, the ESP32+LoRa sensor assembly mounted at ground level, and four environmental sensors (moisture, temperature, pH, gas). **Gateway:** A single solar-powered LoRa gateway with WiFi/4G uplink positioned at the field edge, covering all 16 nodes within a 600 m radius (well within SF7 range of 1.2 km). **Data Resolution:** At 10-minute duty cycles with 82% TX success rate, each node generates approximately 118 data points/day, yielding 1,888 spatiotemporal measurements/day across the field. This density enables detection of localised anomalies (e.g., a drainage failure affecting 2-3 nodes) within 20 minutes of onset.

Estimated deployment cost: 16 nodes x \$40 BOM + 1 gateway (~\$80) + installation labour = approximately \$800 total, with zero recurring energy or battery replacement costs. This compares favourably with commercial agricultural sensor networks costing \$2,000-5,000 for equivalent coverage with battery-dependent nodes.

6. Conclusions

This paper presented a simulation-based co-design framework for a microbial fuel cell energy harvesting system, ESP32-based embedded sensor node, and two-model AI layer for autonomous environmental monitoring. The framework enables rigorous pre-prototyping evaluation using exclusively literature-calibrated parameters. Key findings are:

- * The proposed AI-based adaptive duty-cycle management reduces average energy consumption by 47.5% compared to a fixed always-transmit baseline (0.413 vs 0.787 mJ/cycle), enabling self-sustaining operation within the MFC power envelope.
- * Isolation Forest anomaly detection achieves F1-score = 0.895 +/- 0.021 (5-fold CV, ROC-AUC = 0.932) without requiring labelled training data, making it directly deployable on new installations.
- * LoRa SF7 (41.2 ms ToA, 16.2 uJ per packet) is the optimal default spreading factor for sub-1.2 km links; SF9 provides the best range-energy trade-off for emergency transmissions.
- * The full AI inference pipeline (Isolation Forest ~115 KB + Random Forest ~45 KB, ~12 ms inference latency) runs within the 520 KB SRAM constraint of the ESP32, consistent with TinyML

deployment guidelines [19,20].

- * Parametric robustness testing confirms energy-positive operation in 81.5% of +/-30% parameter variations, demonstrating graceful degradation under adverse conditions.

- * The open-source Python simulation framework enables reproducible design-space exploration before hardware commitment.

Future work will prioritise: (1) benchtop validation using a laboratory soil MFC or programmable power supply emulator to compare measured vs. simulated energy budgets; (2) replacement of the rule-based Random Forest with reinforcement learning for online policy adaptation; (3) concept drift mitigation through sliding-window retraining for long-term field deployments; and (4) multi-node scalability evaluation with LoRa collision modelling and SF allocation optimisation.

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Data Availability Statement:

All simulation code, firmware, and dashboard source files are openly available at <https://github.com/kahyakursat1/mfc-aiot-sensor> (accessed 28 April 2026). Data generated during this study are contained within the paper.

Conflicts of Interest:

The authors declare no conflict of interest.

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